

MarsadLab at PalmX Shared Task: An LLM Benchmark for Arabic Culture and Islamic Civilization

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Abstract

This paper presents our submission to the PalmX 2025 Shared Task on Arabic cultural and religious knowledge comprehension. We focus on training large language models capable of representing domain-specific cultural and religious knowledge in Arabic. Our approach leverages parameter-efficient fine-tuning of the instruction-tuned Qwen2.5-7B model using Low-Rank Adaptation (LoRA). To address the challenges of limited training data, we apply quantization-aware fine-tuning with 4-bit precision, enabling efficient adaptation under constrained resources. The model is further aligned with the multiple-choice evaluation format to enhance task-specific reasoning. Without relying on external data augmentation, our system achieves competitive performance across both the *Arabic General Culture* and *Islamic Culture* subtasks, demonstrating the effectiveness of targeted fine-tuning for enriching cultural and religious knowledge representation in LLMs. On the blind test sets, our systems ranked 7th and 4th in the cultural and Islamic subtasks, respectively. To ensure reproducibility, we make our full codebase and experimental configurations available at <https://github.com/rafiulbiswas/PalmX>.

1 Introduction

Culturally aware language technologies are essential for high-stakes applications—education, public services, healthcare, and content moderation—where responses must be accurate, respectful, and contextually appropriate. In Arabic settings, a lack of cultural and religious grounding can lead to biased or inappropriate outputs, partly due to the predominance of Western-centric training data in large language models (LLMs) (Ayash et al., 2025; Alwajih et al., 2025b). Addressing this gap requires models that can represent and reason over Arabic cultural heritage and Islamic knowledge, as well as standardized evaluations that make such competence measurable (Sadallah et al., 2025a).

To advance cultural and Islamic capabilities in Arabic-centric LLMs PalmX 2025 shared task offered two subtasks—*General Culture* and *Islamic Culture*—using multiple-choice (MCQ) datasets in Modern Standard Arabic (MSA) (Alwajih et al., 2025a). These subtasks probe models’ ability to reason about customs, cuisine, history, and Islamic practices, providing a focused testbed for culturally grounded reasoning in Arabic.

Developing such capabilities is challenging. Beyond data imbalance, Arabic presents diglossia, rich morphology, and strong context dependence, all of which complicate knowledge representation and question answering (Hasan et al., 2025). Practical constraints—limited labeled data and domain-specific MCQ formats—further motivate resource-efficient adaptation strategies.

We adapt an instruction-following LLM to these subtasks using parameter-efficient fine-tuning. Concretely, we fine-tune the 7B-parameter Qwen2.5-Instruct (Team, 2025) with Low-Rank Adaptation (LoRA) (Hu et al., 2022) on the official PalmX training sets (Alwajih et al., 2025a), enabling effective domain adaptation under modest compute. At inference, we employ prompt-based strategies to inject expert priors and enforce output constraints (e.g., instructing the model to act as an “expert in Arabic culture and Islamic studies” and to output only the option letter). Empirically, careful prompt design yields consistent but modest gains in MCQ accuracy; closing the remaining gap will likely require richer cultural grounding and more structured supervision. To summarize, our contributions include:

- We adapt Qwen2.5-7B-Instruct to Arabic cultural and religious knowledge using Low-Rank Adaptation (LoRA) with 4-bit quantization-aware fine-tuning, achieving effective domain specialization under modest computational budgets.
- We introduce inference-time instruction templates and output-space constraints that align the

model with the multiple-choice setting (expert prior + option-letter output), yielding consistent accuracy gains without additional supervision.

- Our system attains resonable performances on PalmX 2025 *General Culture* and *Islamic Culture*, ranking 7th and 4th on the blind test sets, respectively, without recourse to external data augmentation.
- We provide a concise pipeline demonstrating that low-compute PEFT can reliably enrich cultural/religious knowledge in Arabic LLMs.

2 Related Works

Benchmarking language models for Arabic has progressed along two complementary lines: inclusion within multilingual suites and dedicated evaluations of large language models (LLMs) for Arabic. Early efforts commonly incorporated Arabic into broad benchmarks such as XGLUE, XTREME, XTREME-R, GEM, and Dolphin, covering a spectrum of tasks that emphasized classification (e.g., natural language inference), sequence labeling (part-of-speech tagging, named entity recognition), and generation (summarization) (Liang et al., 2020; Hu et al., 2020; Ruder et al., 2021; Gehrmann et al., 2021; Nagoudi et al., 2023). More recent work has turned to Arabic-focused LLM assessment, evaluating standard and Arabic-centric models on task suites and datasets (Sengupta et al., 2023; Khondaker et al., 2023; Abdelali et al., 2024; Dalvi et al., 2024), probing the effects of prompting in native (Arabic) versus non-native (English) languages (Kmainasi et al., 2025), and extending analyses to multimodal settings (Alwajih et al., 2024; Das et al., 2024).

Within cultural evaluation, prior studies quantify representational bias in entity mentions toward Western versus Arab contexts (Naous et al., 2024), assess cultural alignment using constructs from the World Values Survey (AlKhamissi et al., 2024), and introduce culture-aware diagnostic and QA resources (Arora et al., 2024; Myung et al., 2024; Alam et al., 2025). Complementing these efforts, Arabic-focused benchmarks have begun to appear: ARADICE targets dialect comprehension and cultural QA (Mousi et al., 2024), while other resources probe cultural values and regional knowledge via translated survey instruments and Wikipedia-derived questions (Al-Matham et al., 2025). Despite these advances, converging evidence indicates that general-purpose LLMs still un-

derperform on culturally grounded reasoning and Arabic commonsense, underscoring the need for benchmarks, resources, and model-development methods explicitly tailored to Arabic cultural and dialectal contexts (Sadallah et al., 2025b; Yakhni and Chehab, 2025; Qian et al., 2024).

PalmX 2025 (Alwajih et al., 2025a) advances this research area with a curated, competition-driven evaluation of Arabic cultural capabilities in Modern Standard Arabic, spanning *General Culture* and *Islamic Culture*. QAs are designed to cover all Arab countries and key Islamic concepts, providing a focused MCQ testbed and strong baselines (e.g., *NileChat-3B*) (Mekki et al., 2025; Alwajih et al., 2025b). Our work aligns with this direction by adapting an instruction-tuned LLM to PalmX via parameter-efficient fine-tuning and expert-persona prompting, and by analyzing remaining performance gaps relative to culturally trained baselines.

3 Dataset

PalmX 2025 provides two Modern Standard Arabic (MSA) multiple-choice (four-option) datasets that target complementary facets of cultural knowledge.

Task 1: General Culture comprises 4,500 questions spanning Arab culture across 22 countries, with official splits of 2,000 training, 500 development, and 2,000 test items.

Task 2: Islamic Culture contains 1,900 questions focused on Islamic cultural knowledge, split into 600 training, 300 development, and 1,000 test items. All experiments in this work use the organizers’ official splits without external data augmentation.

Both subtasks follow a consistent data distribution structure, previously unseen questions for blind testing, with accuracy serving as the primary evaluation metric (see in figure 1).

4 System Overview

We experimented with different open sources LLM such as NileChat-3B (Mekki et al., 2025), LLaMA3.1 8B (Touvron et al., 2023), Fanar-1-9B-Instruct (Team et al.) and Qwen2.5-7B-Instruct (Team, 2025). Qwen2.5-7B-Instruct outperformed over other LLM and so we adapt Qwen2.5-7B-Instruct to Arabic cultural understanding via parameter-efficient fine-tuning with Low-Rank Adaptation (LoRA) (Hu et al., 2022).

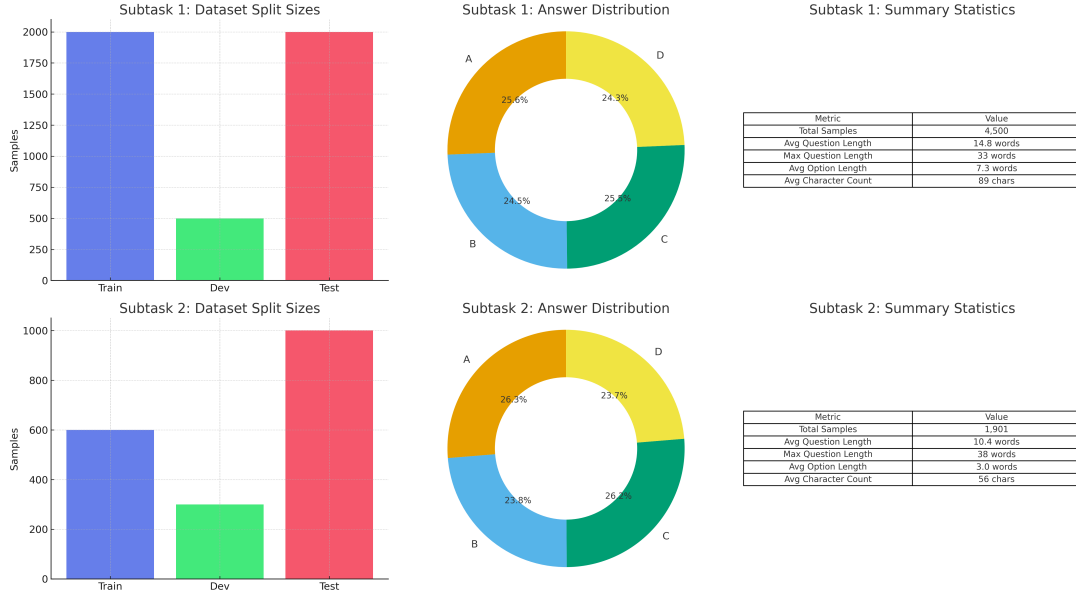


Figure 1: Dataset statistics in two subtasks.

4.1 Training Methodology

Prompting and supervision: We formatted training examples using a structured instruction-following template for Arabic cultural question-answering. Each instance comprises a system message, the user turn containing the question and the four labeled options, and an assistant turn with *only* the correct option letter. We implement this using the model’s native chat template markers (`<|im_start|>` / `<|im_end|>`) to delimit turns. Explicitly constraining the target to the option letter suppresses verbosity, improves label consistency, and simplifies answer extraction at evaluation time; supervision is via standard next-token cross-entropy over the assistant turn.

Optimization setup: We train for three epochs (selected via development-set performance) with a learning rate of 2×10^{-4} and a linear warmup of 100 steps. We use an effective batch size of 16 via per-device batch size = 4 and gradient accumulation $\times 4$. Mixed precision uses bfloat16 where supported (falling back to fp16), and the maximum sequence length is 512 tokens, which comfortably covers all MCQ contexts in our data.

Memory efficiency: To enable fine-tuning on commodity GPUs, we combine 4-bit NF4 quantization of the base weights with gradient checkpointing, trading additional compute for a reduced activation footprint. In practice, this configuration supports single-GPU training with ~ 8 GB of memory. During preprocessing, we tokenize in mini-batches (size 100) to avoid holding the entire

tokenized corpus in memory, and we periodically release cached CUDA memory to mitigate fragmentation during longer runs.

Multi-task adapters: For the two PalmX subtasks, we train separate LoRA adapters on the same quantized backbone to avoid negative transfer across cultural domains while retaining a unified deployment artifact. The *General Culture* adapter is fine-tuned on $\sim 2,000$ instances, and the *Islamic Culture* adapter on ~ 600 instances. This modular design permits task-specific specialization and lightweight “hot-swapping” at inference time without reloading the base model.

4.2 Ablation Study

To better understand the contribution of different components in our system, we conducted comprehensive ablation experiments examining the impact of LoRA hyperparameters, and prompt engineering choices.

LoRA Rank Analysis: We investigated the effect of LoRA rank on model performance and computational efficiency. Table 1 presents results for different rank configurations while keeping other hyperparameters constant ($\alpha = 32$, dropout=0.1).

The results reveal a clear performance improvement from rank 4 to 16, with diminishing returns beyond rank 16. Our chosen rank of 16 represents the optimal balance.

Target Module Selection: We evaluated different combinations of target modules for LoRA adaptation. Table 2 shows the impact of different mod-

Rank	Task 1 (%)	Task 2 (%)	Params (M)	Time (h)
4	63.2	69.8	5.24	2.1
8	65.8	72.1	10.49	2.4
16	67.6	74.1	20.97	3.0
32	67.9	74.3	41.94	3.8

Table 1: Effect of LoRA rank on performance on test dataset

ule combinations. Targeting all projection matrices yields the best performance, with attention modules alone outperforming Feed-Forward Neural (FFN) Network modules, suggesting that adapting attention patterns is more crucial for cultural understanding.

Target Modules	Task 1 (%)	Task 2 (%)
Attention (q, v)	64.3	70.2
Attention (q, k, v, o)	66.1	72.5
FFN only	63.7	69.4
All (Attn + FFN)	67.6	74.1

Table 2: Performance of different target module configurations

Prompt Engineering Variations: We tested several prompt variations to identify the most effective format for Arabic cultural questions.

Prompt Strategy	Task 1 (%)	Task 2 (%)
English system	64.7	71.2
Arabic system	66.3	72.8
Expert framing	67.6	74.1
Expert + few-shot	66.9	73.5

Table 3: Effect of prompt engineering strategies

The expert framing prompt that positions the model as "an expert in Arabic culture and Islamic studies" yields the best results. Adding chain-of-thought or few-shot examples slightly decreased performance.

5 Result

In Table 4, we report the performance of different models for both tasks before and after fine-tuning. Across both subtasks, performance varies markedly by model and adaptation strategy. The fine-tuned Qwen2.5-7B-Instruct yields the strongest overall results, attaining (67.55%) accuracy on *Task 1: General Culture* and (74.13%) on *Task 2: Islamic*

Culture. Fine-tuning provides substantial improvements for all models except Fanar 7B on Task 2, with gains ranging from 7.8 to 12.2 percentage points on Task 1. Notably, Qwen2.5-7B demonstrates the most consistent improvement, gaining 7.75 points on Task 1 and 8.73 points on Task 2. Relative to the task with top ranked system (72.15%) and (84.22%), respectively, this places our best system within (4.60) percentage points on General Culture and (10.09) points on Islamic Culture, indicating substantial room for improvement, especially for the latter.

A cross-task comparison reveals a general trend of improved accuracy on the Islamic subtask after fine-tuning. For Qwen2.5-7B, the gain from Task 1 to Task 2 is (+6.58) percentage points. The NileChat-3B baseline is comparatively stable at ($\approx 64\%$) on both tasks after fine-tuning, while Llama 3.1 8B-Instruct exhibits a modest uplift over this baseline on Islamic Culture (about (+4.9) points). An exception to the broader trend is Fanar 7B, which performs competitively on General Culture (66.0%) but declines on Islamic Culture (62.4%) compared to its baseline performance (49.6%), suggesting that while fine-tuning improves its general performance, domain- or data-mismatch effects persist that merit further analysis.

These results demonstrate three key observations. First, parameter-efficient fine-tuning confers clear benefits over off-the-shelf models for culturally grounded question answering in Arabic, with consistent improvements observed across most model-task combinations. Second, the effectiveness of fine-tuning varies by model architecture and task domain, as evidenced by the differential improvements across models. Third, the persistent gap to the subtask best scores—particularly on Islamic Culture—highlights the difficulty of capturing nuanced, domain-specific knowledge and the need for richer supervision and/or targeted knowledge integration beyond instruction tuning alone.

Model	General Culture		Islamic Culture	
	Before fine tuning(%)	After fine tuning(%)	Before fine tuning(%)	After fine tuning(%)
NileChat-3B	52.30	64.50	51.80	64.00
LLaMA3.1 8B	58.40	65.90	61.70	69.20
Fanar 7B	54.20	66.00	49.60	62.40
Qwen2.5L-7B	59.80	67.55	65.40	74.13

Table 4: Performance comparison of language models on test dataset before and after parameter-efficient fine-tuning. All scores represent accuracy percentages.

Computational efficiency. Our approach is computationally lightweight: training *Task 1* (2,000 samples) completes in approximately three hours on a single NVIDIA RTX 3090; with 4-bit quantization, fine-tuning fits within 8 GB of GPU memory. At inference, throughput is about ~ 2 s per question on GPU and ~ 8 s on CPU. The resulting LoRA checkpoint occupies ~ 1.2 GB, compared to ~ 15 GB for the full model,

6 Error Analysis

To better understand the limitations and failure modes of our fine-tuned models, we conducted a comprehensive error analysis on a stratified sample of 200 incorrect predictions from our best-performing model (**QWEN2.5L-7B**). Our analysis reveals distinct error patterns across tasks: for General Culture, the primary failure modes include factual knowledge gaps (42%), cultural context misunderstanding (28%), and ambiguous question interpretation (18%). For Islamic Culture, errors predominantly stem from religious text interpretation challenges (35%), difficulty handling sectarian variations (24%), and historical timeline confusion (21%).

When comparing errors across tasks, we also observed common problems such as relying on surface-level patterns instead of deeper understanding, showing overconfidence in culturally ambiguous cases, and favoring Western or standardized views over regional cultural perspectives. These findings suggest that while parameter-efficient fine-tuning improves performance, the models still face challenges in handling complex cultural reasoning that requires deeper context and sensitivity to local variations.

7 Conclusion

This paper presented MarsadLab’s approach to the PalmX 2025 shared task on Arabic Islamic and Cultural understanding. Through parameter-efficient fine-tuning using LoRA adaptation of Qwen2.5-7B-Instruct, we achieved competitive performance across both tasks. Our work demonstrates that parameter-efficient methods can effectively adapt LLMs for culturally-nuanced tasks without requiring extensive computational resources. By training only 0.27% of model parameters through LoRA while employing 4-bit quantization, we reduced memory requirements by approximately 75% compared to full fine-tuning, making our approach ac-

cessible to researchers with limited GPU resources. Future work includes investigating low-compute and minimal-data regimes for such tasks.

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Appendix

Model Architecture

We employ Qwen2.5-7B-Instruct, a 7.61B-parameter causal LLM comprising 28 transformer layers with Grouped Query Attention (GQA) and a context window of up to 131,072 tokens. For our setting, we use the instruction-tuned variant—optimized to follow complex prompts via supervised fine-tuning and RLHF. To reduce memory footprint, the base model is loaded with 4-bit NF4 quantization (with double quantization) while retaining bfloat16 compute through bitsandbytes; we enable k-bit training using `prepare_model_for_kbit_training`. Tokenization relies on the Qwen tokenizer with right padding; when a pad token is not defined, we map `<pad>` to `<eos>`.

LoRA Adaptation Strategy

To specialize both attention patterns and intermediate representations for culturally grounded reasoning, we attach LoRA adapters to the attention projections `q_proj`, `k_proj`, `v_proj`, and `o_proj`, as well as to the feed-forward projections `gate_proj`, `up_proj`, and `down_proj`. After development-set tuning, we adopt a rank $r = 16$, scaling $\alpha = 32$, dropout = 0.1, and no bias, a configuration that yields approximately **20.97M** trainable parameters ($\approx 0.27\%$ of the base model). In practice, this supports single-GPU fine-tuning with ~ 8 GB of memory while preserving sufficient capacity for the target tasks.

Hyperparameters

After empirical evaluation on the development set, we selected the following LoRA hyperparameters:

- **Rank (r):** 16 - Balancing expressiveness with parameter efficiency
- **Scaling factor (α):** 32 - Controlling the magnitude of LoRA updates
- **Dropout:** 0.1 - Preventing overfitting on the limited training data
- **Bias:** None - Following standard LoRA practice

This configuration results in approximately **20.97M trainable parameters** (0.27% of total model parameters), enabling fine-tuning with only 8GB of GPU memory while maintaining model expressiveness for cultural reasoning tasks.